



Greendzine Technologies team



Greendzine's Infinity charger charging Quark EV

(50,000 square feet) to achieve a 30% increase in productivity. The products are direct plug and play solutions. The company has distributors all over India and also sells directly to its customers.

Despite having different EV designs for each segment, all of the green mobility solutions come with a single charging module that can be used with any other elec-

tric vehicle, be it a two-wheeler or a three-wheeler. The charging modules can be connected in series and parallel to trigger the charging mechanism to charge each other.

"In power electronics, you cannot connect two power-generating converters in series, because

you would get into load balancing issues. We also faced that problem, but we were able to solve it through multiple iterations, and we ultimately used a special switching mechanism called GaN FET. We were able to switch GaN FET switches at a frequency of 1MHz, increasing the power density of the charger," explains co-founder Karthikeyan Sundaram.

Sundaram says that the team overcame the challenge to prevent the switch from bursting or catching fire at that frequency. The company has added this technology along with the cooling system for the charger to its list of IPs. While using immersive cooling technology, the power electronics board is immersed in a water-like solution, which is developed in-house. The whole charging module is compact.

"Our charger is half the size of the ones available in the market but can deliver the same capacity

of power as the ones available in the market. The same charger can be used to charge an electric cycle. Two charging modules can be used to charge a bike. Three can be used to charge autos and cars. More than four or five can be used to charge an electric bus," Sundaram informs.

Incubated under IIIT Bengaluru Innovation Centre, the startup has received an undisclosed amount of funding from Karnataka IT Venture Capital (KITVEN) Fund. The company designs the components and places an order with its manufacturing partners for the same. Once the components are received, it assembles the parts and tests the EVs at the Bengaluru facility.

Greendzine is currently looking for powertrain and ECU components vendors and partners to expand its business. The company has managed to veer its way through inflation and supply chain constraints.

Co-founder Anjan Kumar N. says that they overcame these challenges by resizing their order size for production and re-negotiating vendor contracts. The company is currently running pilots to connect all its EV products onto one single software platform. To ensure that the economy yields more productivity per person, they would need to keep automating their lower-value-addition jobs, so that people can keep moving towards higher-value-addition jobs and solutions like MOPTro.

—Yashasvini Razdan

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A SEMI-BIONIC HAND THAT HELPS THE USERS TO TOUCH AND FEEL

A bionic arm is not the last resort but a new beginning, a second chance at life. Robo Bionics is aiding people having arms disability through their bionic arm, Grippy. The company hopes to make bionic arms affordable for all

While doing M.Tech at IIT Patna, a close friend of Llewellyn D'sa had arm

disability since birth. The problems faced by his friend inspired him to do something for people hav-

ing similar disabilities. On learning about bionic arms, he realised that most prosthetics available in the